Brayden Goldstein-Gelb

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EDUCATION

Boston University, PhD Boston, MA | 2026 – Present
Computer Science, Advisor: Alexander Poremba
- Relevant coursework: Complexity Theory, Polynomial Methods for Algorithms

Brown University, Sc.B, 3.91/4.00 GPA (Magna Cum Laude) Providence, RI | 2021 – 2025
Computer Science (Honors), Mathematics; Honors Thesis: Distributed Quantum Signal Processing
- Relevant coursework: Quantum Mechanics, Quantum Information, Algorithm Design/Analysis, Computer Systems, Cryptography, Machine Learning, Theory of Computation, Linear Algebra, Algebra I-II (Grad.), Real and Complex Analysis

SKILLS

- Programming Languages: Python, HTML/CSS, JavaScript, Java, C/C++, Julia, Pyret
- Technologies: Qiskit, Stim, Git, NumPy, Sklearn, Pandas, TensorFlow, PyTorch, Flask, PostgreSQL, Vue, D3, Node.js

Work Experience

NewGrid Inc. Somerville, MA
Software Engineer II July 2026 – Aug 2026
Software Engineer I Aug 2025 – June 2026
Software Engineer Intern May 2024 – Aug 2024
- Implemented search functionality and performance enhancements for topology optimization algorithms.
- Built an external API and SDK for NewGrid's power grid simulation and analysis software.
- Automated the scraping, processing, and storage of power grid outage data, streamlining data collection for integration with NewGrid's database and enabling real-time monitoring for the service team.
- Developed desktop application features to automate the import and export of grid settings, reducing manual setup time.

SULI Program, Oak Ridge National Lab Oak Ridge, TN
Quantum Computing Research Intern May 2023 – Jul 2023
- Investigated methods for solving constrained optimization problems using QAOA.
- Developed a procedural method to translate constraints into mixing Hamiltonians and proved convergence guarantees.

PUBLICATIONS

Brayden Goldstein-Gelb and Phillip C. Lotshaw. "Convergence Guarantee for Linearly-Constrained Combinatorial Optimization with a Quantum Alternating Operator Ansatz." *ACM Transactions on Quantum Computing* (2026). To appear. arXiv:2409.18829

Brayden Goldstein-Gelb, Kun Liu, John M. Martyn, Hengyun Zhou, Yongshan Ding, Yuan Liu. "COMPAS: A Distributed Multi-Party SWAP Test for Parallel Quantum Algorithms." In Proceedings of the 31st *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '26)*, Volume 2, pp. 426–441. arXiv:2511.23434

LEADERSHIP

Brown University Providence, RI
Grader, Linear Algebra with Theory Jan 2025 – May 2025
Teaching Assistant, Theory of Computation Aug 2024 – Dec 2024
- Created problem sets and example solutions, assisted in grading, held weekly office hours

Wyzant Tutoring Somerville, MA
Math and Computer Science Tutor May 2021 – May 2023
- Over 350 hours of tutoring experience, working with 50+ students ranging from middle schoolers to executives.
- Maintained a five-star rating, helping students build confidence and mastery in Python, Java, Calculus, and Algebra.

PROJECTS

Hackathon Projects: *MIT iQuHack, Brown Quantum Hackathon, Hack@Brown*
- Won 2nd place at Brown Quantum Hackathon
- Completed QuEra challenge at MIT's iQuHack quantum hackathon
- Created a space-themed game for Hack@Brown

Zero-Knowledge Proof for Graph 3-Coloring, *Final Project, Applied Cryptography*
- Implemented an interactive zero-knowledge protocol to prove graph 3-colorability without revealing the coloring
- Built a terminal-based application with verifier/prover roles communicating over TCP sockets.

Quantum Mechanics Proofs Library, *Final Project, Formal Proof and Verification*
- Created a library in the Lean Theorem Prover to model quantum particles, measurements, and multi-particle states.
- Demonstrated the library's capabilities by constructing a formal proof of the no-cloning theorem.

More projects, additional details, and demos available at braydengg.com/projects.